

**INTEGRATED CLIMATE-INTELLIGENT SALT PRODUCTION
ECOSYSTEM: A MULTI-MODAL AI FRAMEWORK FOR
SUSTAINABLE YIELD OPTIMIZATION IN TROPICAL COASTAL
ENVIRONMENTS**

Project ID: 25-26J-431

Project Design Report for Checklist 01

B.Sc. (Hons) Degree in Information Technology, specializing in
software engineering

Department of Software Engineering
Faculty of Computing

Sri Lanka Institute of Information Technology

Sri Lanka

WireFrames

<https://excalidraw.com/#json=A-FveHNYddUQbEE9J4zd,dGpsEz2P3xXflwtT3tYRrA>

Figma Prototype URL

<https://www.figma.com/design/XgBPsME7DaGH24F4g1QDkk/Brinex?node-id=0-1&t=vXdYW3aYRqnODaxp-1>

Feedback on each component.

01. Crystallization Forecasting

- Puttalam Salt Society confirmed they mostly produce standard table salt, so the main need is total production forecasting, not per crystal type.
- Users preferred having Tamil and Sinhala language support.

02. Intelligent Planning Engine

- The landowners mentioned the hardship of getting the best offers from sellers. A unified platform would be very useful and easy for their decision-making.
- User requested Language support.
- Requested to simplify the dashboard since neither the landowners nor the sellers are familiar with technology.

03. Computer Vision Quality Control System

- Positive: Users praised detection accuracy (matching human inspectors), clear real-time visualization, and automated batch tracking that reduced manual record-keeping.
- Improvements: Future enhancements include camera auto-calibration, GPU acceleration for high-throughput, and PDF/Excel export for compliance.

04. Federated Learning for Waste Valorization

- Users valued the privacy-preserving approach and improved prediction accuracy as more participants joined, making it well-suited by for cooperative societies like Puttalam Salt Producers.
- Future Enhancements: Simplified waste value visualization